

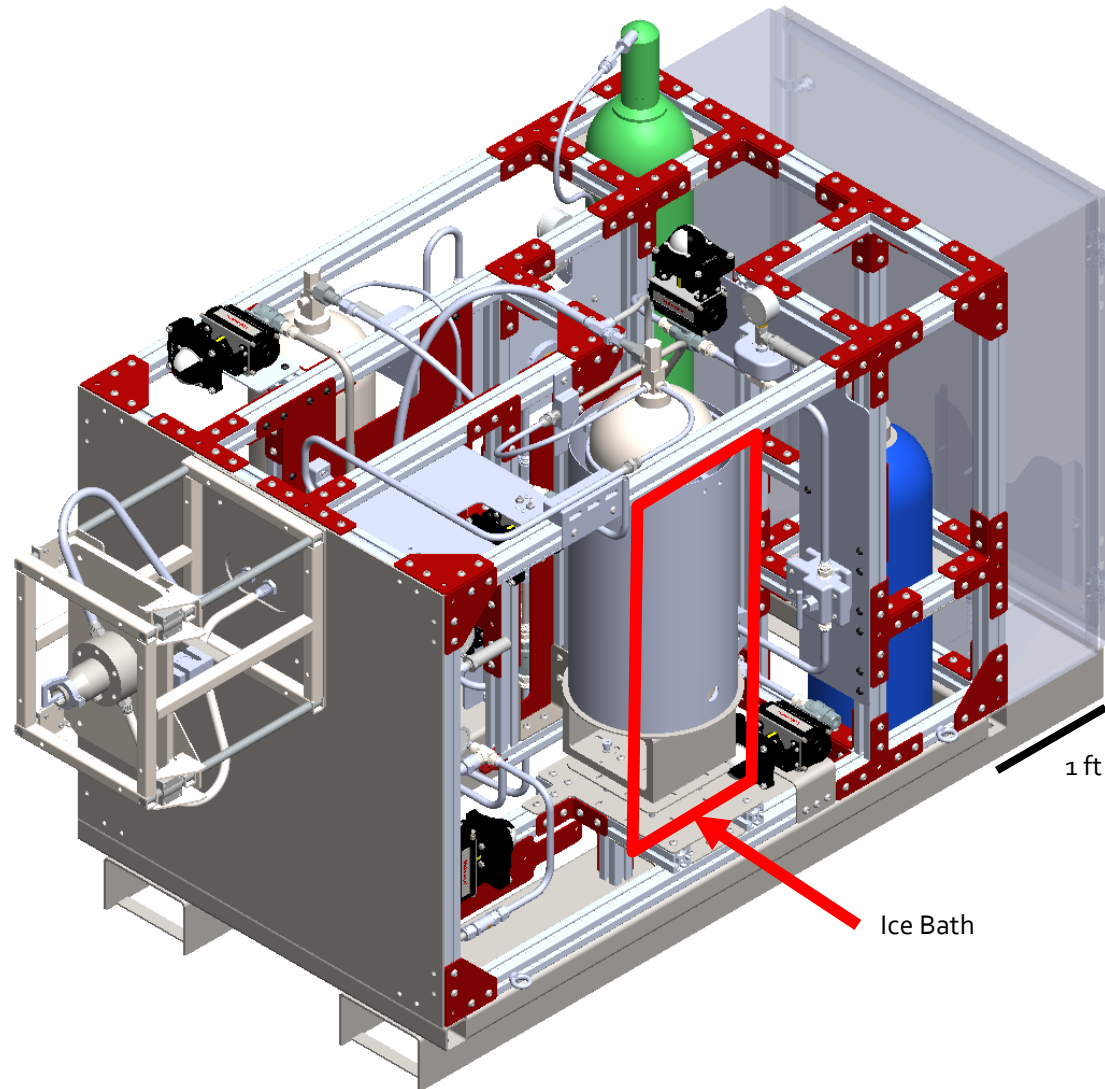
Engineering Project Portfolio

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Rose-Hulman ME 29'

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Rose-Hulman Liquid Fueled Rocket



CAD model of test stand

Background: I'm the Structures Lead on the Rose Propulsion Laboratory team, competing in the College Propulsive Lander Challenge. Last year, we built a test stand from scratch for our first engine prototype, which uses an N_2O -IPA fuel blend.

My Objective: At a feed system pressure of 700 psi, the N_2O oxidizer must be kept below 18°C to stay in the liquid phase. Our engine is designed for the oxidizer to enter the combustion chamber as a liquid, so we needed a cooling solution for our nitrous oxide to allow us to conduct tests in warm conditions.

Constraints:

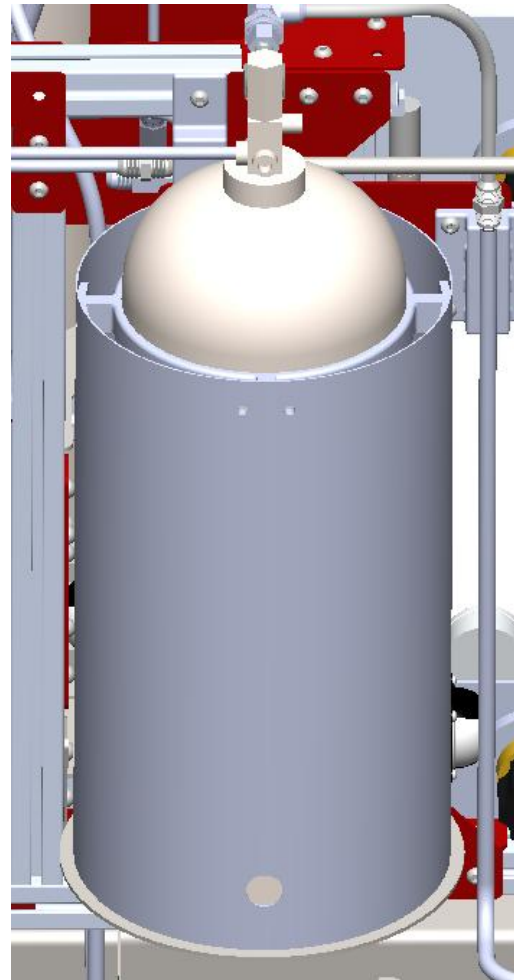
- Very limited space (without moving feed system components)
- Must have sufficient capacity to initially cool N_2O tank and keep it cool for an entire day of testing
- Must allow for easy N_2O tank swaps
- Can not corrode
- Must accommodate loadcell mounting points at base
- Should be manufacturable

Nitrous Oxide Cooling Tank

N ₂ O Saturation Curve				
Temp (C)	Pressure (psia)	Density (kg/m ³)	Volume (m ³ /kg)	Phase
3.917	500.0	886.6	0.00112	liquid
11.39	600.0	843.7	0.00118	liquid
17.99	700.0	800.0	0.00125	liquid
23.90	800.0	753.0	0.00132	liquid
29.25	900.0	697.7	0.00143	liquid
34.10	1000	616.3	0.00162	liquid

- N₂O saturation curve from NIST Chemistry WebBook
- Feed system designed for an operating pressure of ~700 psi, so the ice bath must keep N₂O below 18 °C

Completed cooling tank



CAD model of the cooling tank and N₂O tank mounted into rocket test stand

Engineering Process:

- Determined required cooling capacity under worst-case ambient temperature scenario and sized cooling tank appropriately
- Selected slip-rolled 304 stainless steel for its corrosion resistance, strength, and relative ease of welding
- Modeled cooling tank and centering ring in SolidWorks
- Implemented design changes suggested by peers during design review, such as switching centering ring mounting holes to slots to account for manufacturing variations
- Cultivated new sponsor relationship to manufacture cooling tank, and provided said sponsor with relevant engineering drawings and CAD files

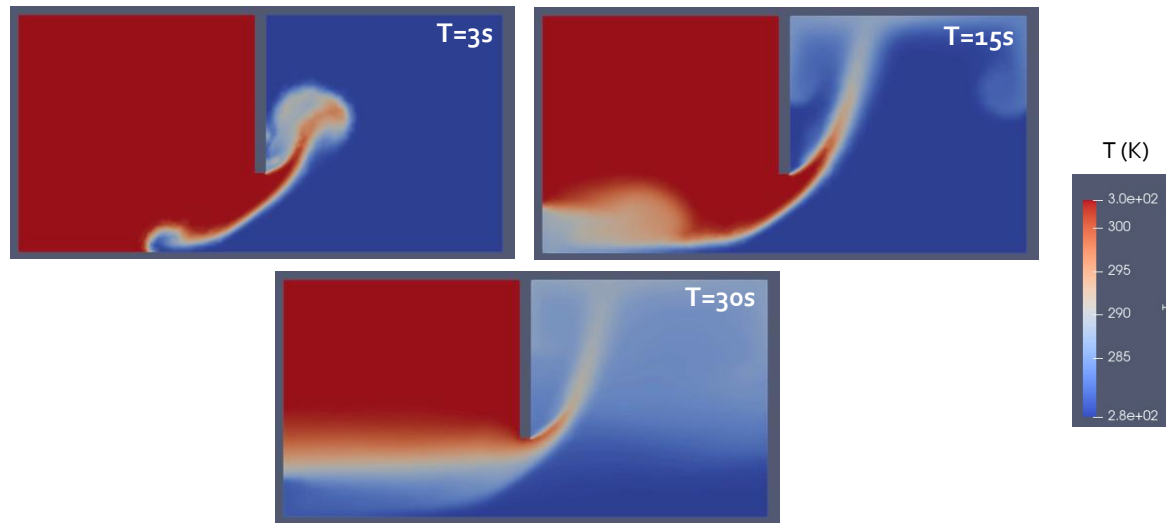
Outcome:

- Conditioned our nitrous oxide to required temperatures during cold flow and hot fire attempts

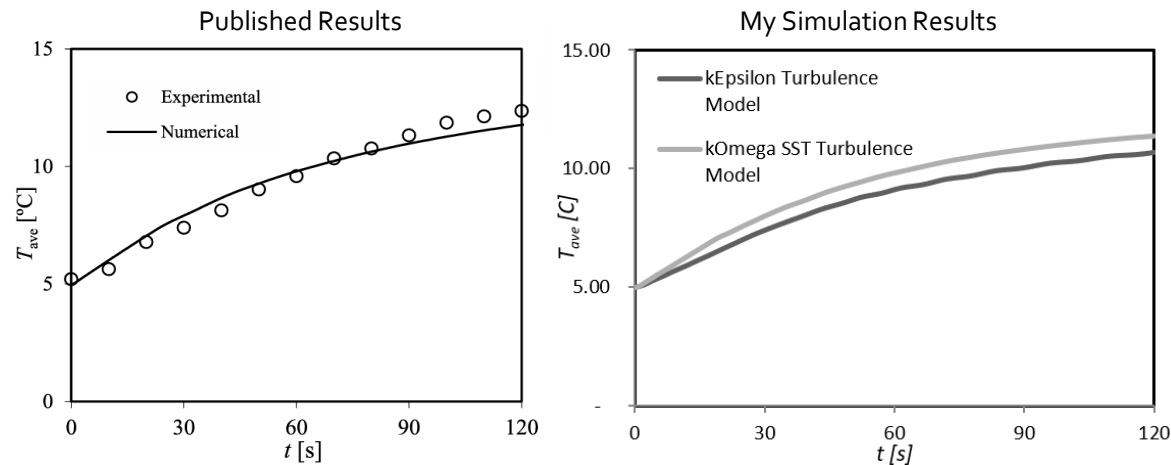
Skills Developed:

- Thermal Analysis
- SOLIDWORKS Sheet Metal
- Design for Manufacturability

Internship: Simulating Air Infiltration Using CFD



Slices of simulation showing temperature distributions at indicated times. The left chamber was initialized at a temperature of 30°C , the right chamber was initialized at 5°C .



Objective: Creative Thermal Solutions Inc. (summer 2026 internship) asked me to replicate published ANSYS CFX results showing doorway air infiltration using the open-source C++ CFD program Open FOAM.

Engineering Process:

- Completed Open FOAM tutorials and reviewed the simulation's setup by a former PhD research engineer, who was the only person at the company who knew how to use OpenFOAM
- Created and imported meshes using SALOME
- Configured initial temperatures and CO₂ tracer gas mass fractions using field setting tools
- Set up initial steady-state simulation to rapidly find and fix initial and boundary conditions issues and then switched to transient simulation
- Added temperature and CO₂ concentration probes to log data and ultimately create my own figures to compare with the paper's results

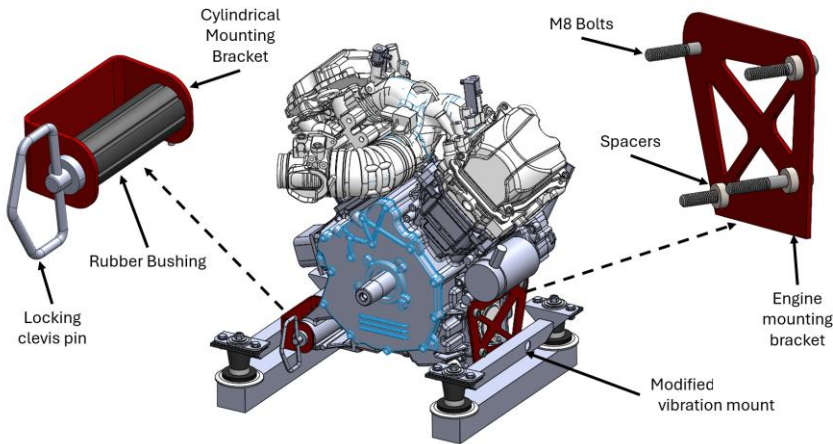
Outcome:

- Successfully simulated 120 seconds of transient, buoyancy-driven flow and recorded data closely matching the simulation results from the paper
- Developed sufficient OpenFOAM knowledge to train CTS engineers on the software

Skills Developed:

- Understanding of OpenFOAM and CFD
- Creating high-quality meshes using SALOME
- Visualizing results and exporting data in ParaView

Design Project: Hovercraft Engine Mounting System



CAD assembly of our mounts with the engine

Technical specifications and metrics for our engine mount

Technical Specifications		Met?
TS-1a	A single engine mount weighs less than 51 lbs.	Yes
TS-1b	A single engine mount weighs less than 20 lbs.	Yes
TS-2	The engine mount supports a static load of 190 lbs.	Yes
TS-3	The engine mount supports a torque of 74 ft.-lbs.	Yes
TS-4a	The engine mount supports a crash load of 6G.	Yes
TS-4b	The engine mount supports a crash load of 3G.	Yes
TS-5a	The engine mount releases the engine in under 10 operations.	Yes
TS-5b	The engine mount releases the engine in 5 or under operations.	No
TS-6	The engine mount must be painted to the required thickness (240 µm) per ISO 12944-5 (Class C4-high atmospheric corrosivity, High (h) durability target) for abrasive blasted steel and a marine environment.	Yes
TS-7	We will use the proven, existing vibration dampeners.	Yes
TS-8	The engine mount can be manufactured in under 10 operations.	Yes
TS-9	When stationary, the driveshaft of the engine should be aligned with the transmission input shaft within a ±3-degree tolerance when measured in the X,Y, and Z.	Yes
TS-10	The mount will be made from materials with a high enough yield strength to prevent the mount from deforming under continuous load.	Yes

Finished prototype components



Plasma cut mount components ready for press brake and mill

Objective: A hovercraft manufacturer challenged my ASME freshman design team to design and prototype an engine mounting system for a new engine model

Engineering Process:

- Discussed system requirements with manufacturer’s engineering team, saw previous engine mounts and learned what worked well
- Rapidly prototyped our initial ideas as a group and shared design with class and engineering team
- Designed mounting system in CAD and created physical prototype using a plasma cutter, press brake, and mill
- Completed design calculations to ensure that the mounts would survive a 5G crash and not fail due to fatigue

Outcome:

- Provided a physical prototype ready to be welded to the hovercraft’s frame
- Demonstrated that our mount could release the engine in under 10 operations

Skills Developed:

- Leadership and Task Delegation
- Design for Serviceability
- Conforming designs to ISO standards